

Software Design Document

Project Name: Ablerr

1. Introduction

Ablerr is a web-based reading assistance application designed for students with cognitive disabilities, especially ADHD. The purpose of this system is to help users improve focus while reading by highlighting one line of text at a time and dimming surrounding content. The application also provides reminder features to support consistent study habits.

2. System Overview

The system is designed for students who experience difficulty concentrating on large blocks of text. Ablerr allows users to log in, upload or input reading material, and start a reading session with a focus mode feature. The system also enables users to create and manage reminders.

The main goal of the system is to provide a simple, accessible, and effective reading support tool.

3. System Architecture

Ablerr follows a three-tier architecture:

Presentation Layer (Frontend):

Implemented using Angular. It provides the user interface, including login, dashboard, reading page, and reminder page.

Application Layer (Backend):

Implemented using Python. It handles user authentication, reading session management, focus mode logic, and reminder processing.

Data Layer (Database):

Implemented using PostgreSQL. It stores user data, reading materials, and reminders.

This architecture ensures separation of concerns and allows easy maintenance and scalability.

4. Functional Requirements

The system shall:

Allow users to register and log in securely.

Allow users to upload or input reading text.

Allow users to start a reading session.

Highlight the current reading line and dim surrounding text.

Allow users to create, edit, and delete reminders.

Store user data and session data in the database.

5. Non-Functional Requirements

The system should be easy to use and accessible for users with cognitive disabilities.

The system should respond within 2 seconds under normal conditions.

The system should ensure basic data security for user information.

The system should be maintainable and modular.

The system should be compatible with modern web browsers.

6. User Interface Design

The system includes the following main pages:

Login Page: Allows users to enter email and password.

Dashboard: Provides navigation to reading sessions and reminders.

Reading Page: Displays text with focus mode enabled (highlight + dim).

Reminder Page: Allows users to manage reminders.

The UI is designed to be simple and minimize distractions.

7. Testing Overview

The system will be tested using functional and black-box testing methods. Test cases will verify core features such as login, reading session functionality, focus mode, and reminder management. The goal is to ensure that all features work correctly from a user perspective.